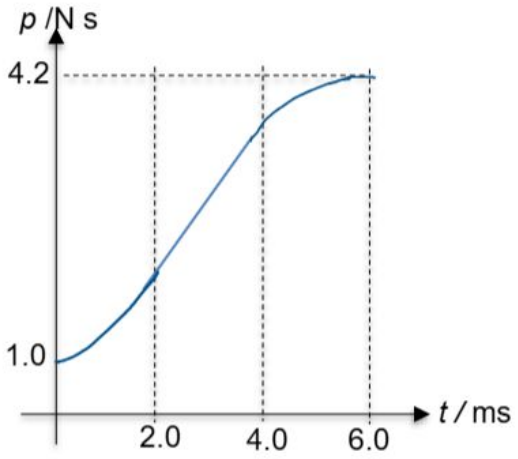
3	(a)	The <u>total momentum of a system</u> of objects <u>remains constant</u> provided <u>no resultant external force</u> acts on the system. [B1]		
	(b)	Impulse is given by the area under the force-time graph (for a body).		[B1]
	(b)	(i)	Area under the *F-t* graph for 6.0 s = 0.5(0.002+0.006)(800) = 3.2 Ns	[C1]
			$\Delta p = p_{final} - p_{inital}$ $p_{final} = \Delta p + mv_{inital}$ $= 3.2 + 0.20(5.0)$ [C1] $= 4.2$ Ns [A1]	
		(ii)	 **Fig. 3.2** 1m for correct shape showing that for the first 2.0 s, (zero gradient initially and then) gradient increase/steeper, for the next 2.0 s, straight sloped line with no kinks, for the last 2.0 s gradient decrease/getting gentler (ending with zero gradient). Graph MUST be smooth with no kinks. 1m for labelling of initial *p* and final *p* value	
	(d)	There is a net <u>external force</u> that is acting <u>on the ball by the wall</u> during collision. [M1] Hence by the Principle of Conservation of Momentum, the <u>momentum of the ball is not conserved</u>. [A1]		